Mechanistic Interpretability in Economic Forecasting Models

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Mechanistic Interpretability (MI) refers to the process of reverse-engineering the internal computations of a neural network to understand how specific components (e.g., neurons, attention heads, or layers) contribute to model behavior.

If we have a NN

$$\boldsymbol{f}\boldsymbol{:}\boldsymbol{X} \rightarrow \boldsymbol{Y}$$

and a hidden state or embedding vector $h_l(\boldsymbol{x})$ at layer l for input $\boldsymbol{x}$ then, we aim to identify a function φ such that:

$$\boldsymbol{\varphi(h_l(x))\cong y}$$

So, it captures interpretable features of f from h_l.

What is Mechanistic Interpretability (MI)?

- Growing concerns over the robustness of development and usage of Language Models (LMs) as they become increasingly important in real-world applications
- MI has emerged as a promising technique to close that gap in the field of interpretability of LMs
- **Goal**: Reverse Engineer detailed computations performed by LM into human-understandable algorithms
- **Intuition**: A programmer trying to decode binaries into human-readable source-code

MI Research Areas

a. **Features**: input property encoded in LM activations

- MI aims to interpret LM representations by decoding features encoded in them
- Suppose we have LM from NLP.
- A Feature: Is this word '**x**' a question word (who, what, why...)?
- An Input: "**What** time is the meeting?"
- Model internal state: Somewhere in the network, a neuron activates strongly when "what" is the input.

So, the model has internally learned the *feature* "question word" — even though it was never explicitly told what that means.

b. **Circuits** - computational pathways connecting features model LM as a network

c. **Universality** - do similar features and features exist on other LMs?

Why MI?

- Bottom-up interpretability technique
- Breaks the model down into into smallest units of analysis (neurons)
- Understand these components and then piece them together to understand how they combine to produce model's overall behavior
- Top-down approaches start from the higher-level model and work to explore granular mechanisms

Techniques:

1. Intervention-based, Vocabulary Projection etc.
2. **Probing**:

- investigate whether a pre-defined feature is present in intermediate activations (intermediate computed values) within the model
- Train a shallow (simple) classifier, known as a probe, to predict whether a feature is present in activations
- Indicate correlation, not causation between features and activations

Feature Studies

1. **Targeted**

a. Generate a Hypothesis (human-reliant): does this LM activation (or even this neuron) encode this feature?

b. Verify the Hypothesis: Directly probe LM representation and inquire about presence of targeted feature F using a probing method

- A probe determining the existence of F is trained

2. **Open-Ended**

- Enumerate all features within activations of LM representation

MI in Economic Forecasting Models: Motivation

- Transformers, CNN’s, DNN’s etc. are routinely used as economic forecasting models - predicting GDP growth, currency crises, inflation etc. from indicators
- These models are opaque
- Economic forecasting is high-stakes and understanding model biases are critical before basing decisions or research on them

The Experiment:

- **Guiding Question:** Can we use data on various macroeconomic indicators of a country over the past consecutive 5 years to make a prediction about the probability of it occurring in the next year?

- **Data:**
  - World Bank and HBS Global Crisis Data
  - 8 countries: Australia, Bangladesh, Brazil, Hungary, India, Korea, South Africa, Turkiye - 8 one-hot encodings
  - 5 numeric economic indicators: Inflation, consumer prices (annual %), GDP growth (annual %), Current account balance (% GDP), Official exchange rate (LCU per US$, period average), Total reserves (includes gold, current US$)

Processed Data:

- 248 samples: X[i] and Y[i]
- Each input: a (5, 13) matrix
- Each output: Labels 1 or 0 (crisis or not)

Experimentation:

Run 3 models!

M1: 1st model: Custom transformer model with hard-coded hyperparameters, 1 encoder layer, and the first time step defined as the the first row of each X[i] that collects context through self-attention

- Binary classification: Will there be a currency crisis?
- 20 epochs, validated on valid set

M2: 2nd model: Hyperparameter tuning by grid search to find best set

M3: 3rd model: Hardcoded hyperparameters from model 2, trained on train+valid, and evaluated on valid split

Final Model:

Comparison metrics:

- **Accuracy** (proportion of correct predictions)
- **F1 scores** (harmonic mean of precision and recall — it balances how many positive predictions are correct (*precision*) and how many actual positives were found (*recall*))

Model	Evaluation Set	Accuracy	F1 Score
M1: Baseline	Validation	0.8108	0.5333
M2: Tuned	Validation	0.7838	0.5
M3: Final	Test	0.7027	0.3529

Probing: Targeted Feature Study

Goal:

1. Test which layers “aware” of crisis labels
2. Is the model classifying particular countries as part of a ‘high-risk’ group and more likely to have currency crises?

Workflow:

1. Hypothesis generation: Does this Transformer activation (e.g., CLS token at layer l) encode the is_crisis_year_next feature? Does this Transformer activation (e.g., CLS token at layer l) encode the is_high_risk_country feature?”
2. Verification by directly inquiring LM activations by training a shallow probe

1. **Does this Transformer activation (e.g., CLS token at layer l) encode the is_crisis_year_next feature?**

a. **Input:** CLS token derived from whole X[i]
b. **Target:** crisis label
c. linear probing for targeted feature study
d. extract the first time step from each transformer layer and store: [input_proj_output, layer1_output]
e. track summary of the 5-year sequence and how it evolves as it goes through the model: token acquires context through self-attention as it goes through the layers
f. on each time step, train a logistic regression classifier
g. evaluate how well this probing performs
h. If accuracy/F1 is high at a layer, it means the CLS token at that layer contains a strong crisis signal.

Layer	Accuracy	F1 Score
0.0	0.857143	0.222222
1.0	0.873469	0.45614

Does this Transformer activation (e.g., CLS token at layer l) encode the is_high_risk_country feature?”

a. Similar probing structure: same input, different output
b. Hard-code: Brazil, Turkey, South Africa as high-risk countries
c. Assign binary label based on country (1 if from high-risk group, else 0)
d. Probe model to predict this label from CLS token

Layer	Accuracy	F1 Score
0.0	1.0	1.0
1.0	1.0	1.0

Takeaways:

1. **Transformer layers progressively encode crisis-relevant information**

2. **Country group identity is perfectly retained** — even without training

3. Probing reveals **when and where the model makes decisions**

4. MI gives us tools to understand **biases**, **feature reliance**, and **model logic**

Future Improvements

- **Longer input sequences**: Include more than 5 years of macroeconomic data to capture longer-term trends.
- **Probing at head-level granularity**: Identify which attention heads carry economic or country-specific features
- **Ablation studies**: Remove macro features or country identity to observe changes in prediction confidence

References:

Rai, D. et al. (2025). *A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models*. arXiv.

Reinhart & Rogoff (Global Crises Data)

World Bank Macroeconomic Indicators